

UPDATE Data

Learning Objectives

- Write UPDATE statements to modify existing records
- Use WHERE clauses to target specific rows for updates
- Understand the risks of updating without a WHERE condition

Engage (5-7 minutes)

A student changes their phone number. The school database needs to reflect this new number for the correct student only. The UPDATE command modifies existing records without affecting other data in the table.

Explore (10-12 minutes)

Given a Students table, write an UPDATE statement to change Grade from “B” to “A” for student ID 5. Then try updating without a WHERE clause and observe how all rows are affected. Discuss why this is dangerous in a production database.

Explain (10-12 minutes)

UPDATE modifies existing records in a table. Set the new values using column = value pairs. The WHERE clause filters which rows to update—without it, every row in the table is changed. Always preview affected rows using SELECT with the same WHERE clause before running UPDATE.

Elaborate (8-10 minutes)

A teacher needs to update marks for a student after a re-evaluation. Write an UPDATE statement that changes the Marks column for a specific student identified by their StudentID. Explain the steps to verify the change was applied correctly.

Evaluate (5-8 minutes)

Write an UPDATE statement to change the Department of employee with EmployeeID 12 from “Sales” to “Marketing”. Explain why running UPDATE Employees SET Department = ‘Marketing’ without WHERE would be problematic.